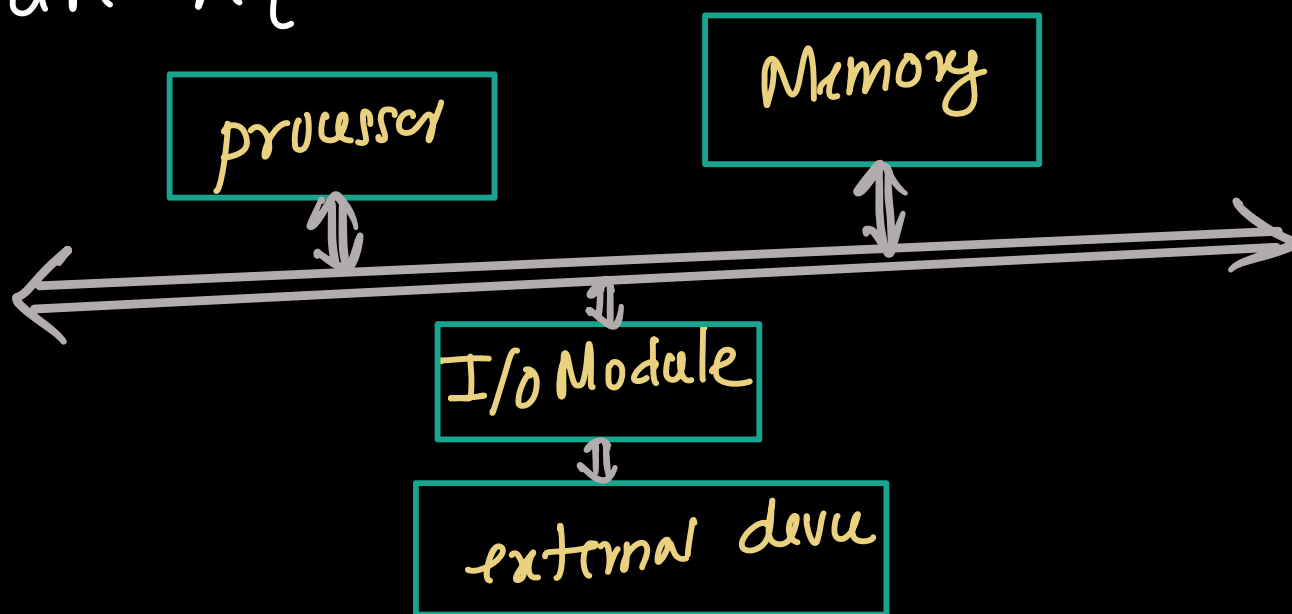


Methods to perform I/O operation (Note)

- I/O operations refers to the process of transferring data between the computer (processor & memory) and external device.
- I/O module is used as interface to manage communication b/w processor and external devices.
- During I/O operation processor issues commands to the I/O module, which communicates with external device and transfer the data between the device and processor.

Need of special I/O techniques

- external devices operate much slower than the processor. Because of this speed difference, special techniques are required to co-ordinate the data transfer.



- Main techniques used for the I/O operations are

- ① Programmed I/O
- ② Interrupt driven I/O
- ③ Direct Memory Access (DMA)

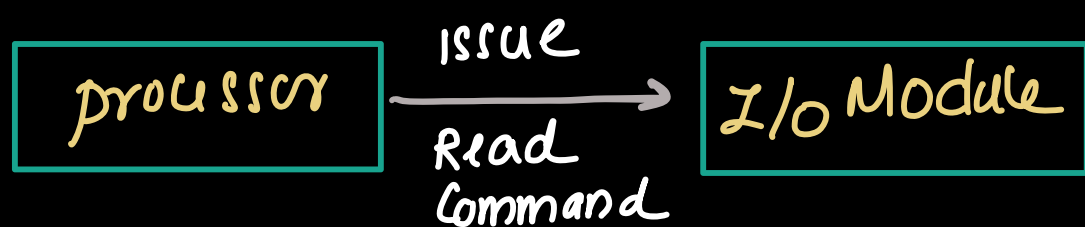
Programmed I/O

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- Programmed I/O is a technique in which the processor controls all I/O operations directly.
- When processor execute an instruction related to input/output, it sends command to the I/O module.
- I/O module performs the requested operation and update status register to indicate the condition of the device
- therefore processor continuously check the status register of the I/O module to determine operation has finished.
(Device is ready)
- This repeated checking of device status is called polling.
- Thus in programmed I/O, the processor remains actively involved in the entire I/O operation.

Eg: We need to read a block of data from I/O device using programmed I/O.

Step 1. processor issue an read command to the I/O module.



- This command instruct the I/O module to obtain data from the external device.
- during this time I/O module send the control signal to external device and external device send the data to external device interface logic.

→ then data is temporarily stored in data register and update the device status in status register. (N)

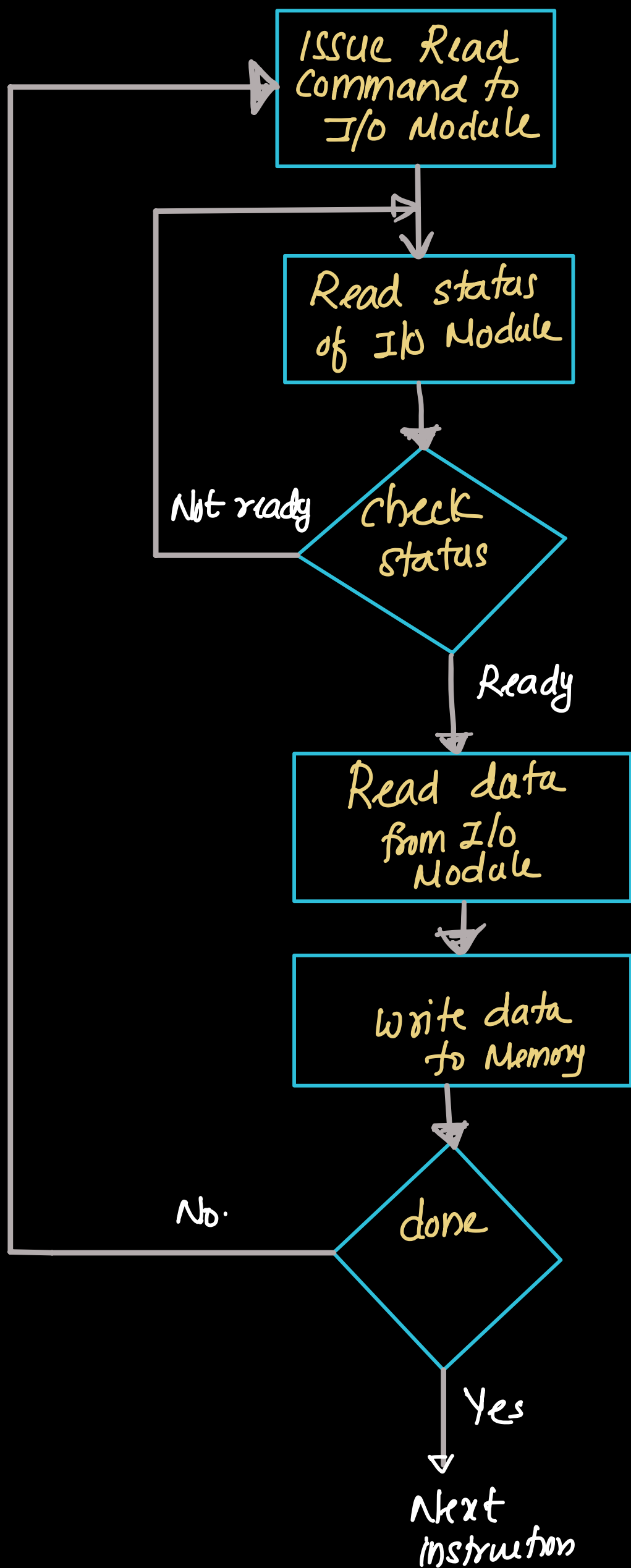
Step-2: After issuing the read command, processor reads status register of the I/O module.
Status Register contains information such as
① device ready
② device busy.

Step 4: Read data from I/O module
— when device become ready, processor reads the data from I/O module through data bus.
— thus data transfer to the processor.

Step 5: Write data into memory.
After receiving the data, the processor store the data into main memory.

Step 6: The processor check whether the entire block of data is transferred or not.
If entire block of data transferred, then processor proceed to execute next instruction.
If more data remains to be transferred, processor return to the first step and repeat the process.

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Disadvantage

- processor must continuously check the device status.
- During that time processor cannot perform other useful tasks. As a result processor time is wasted, making programmed I/O an inefficient technique for data transfer.

Addressing of Memory and I/O devices (Note).

- In a computer system, the processor, main memory and I/O module communicate through a common system bus.
- The system bus consists
 - ① Address bus
 - ② data bus
 - ③ Control bus.
- When the processor needs to access the memory, it places the address of memory location on the address bus. memory unit recognizes this address and transfer the required data through the data bus.
- Similarly when the processor needs to communicate with an I/O device. processor places address of I/O device / add of I/O module register on the address bus. The corresponding I/O module recognize the address and exchange the data through system bus.
- Since memory, processor, I/O devices share the same system bus, processor must be able to distinguish b/w memory address and I/O address. Two addressing methods are used for this purpose.

① Memory mapped I/O

② Isolated I/O.

Memory Mapped I/O.

(2)

- In memory mapped I/O, a single Address space is used for both memory location and I/O devices (I/O module Register).
- Processor treat the status and data register of I/O module as memory locations.
- Therefore processor use the same instruction to access both memory and I/O devices.

- Eg: If the system has have the 10 Address lines.

$$\text{Total No. of Address} = 2^{10} = \underline{\underline{1024}}$$

- First 512 Address are used to address the memory location.

- Remaining 512 Address are reserved for I/O module Registers / I/O devices

eg: 516 → assigned for data Register
517 → assigned for status Register.

→ Eg: - A keyboard is connected to the computer s/m. and processor need to read data from keyboard.

- two memory Add are reserved for the keyboard

517 - Keyboard status Register

516 → keyboard data Register.

program.

① processor load the value 1 into the accumulator

load Ae, 1

② processor store the value 1 into address 517
this initiates the keyboard read operation

load AE, S17

- (3) processor check the status Register (S17) to determine whether the data is ready

load AE, S17

- (4) processor repeatedly check the status Register until the ready bit becomes 1.

load AE, 1

store AE, S17

L1 load AE, S17
Branch if sign = 0 ; L1

- (5) Read the input data
if ready bit become 1, then processor reads the data from data Register

load AE, 1

store AE, S17

L1 load AE, S17
Branch if sign = 0 ; L1
load AE, S16

Advantages

→ Same instruction can be used to access the memory and I/O devices, which allows more flexible and efficient programming.

Isolated I/O (Note)

- In isolated I/O, the address space used for I/O devices is separate from memory address space. Therefore memory and I/O devices have independent address ranges.
- For eg: , with 10 address line, the system may support
1024 memory Address,
1024 I/O addresses.
- In this technique, processor communicates with I/O devices using special I/O instructions such as IN and OUT
IN - to read data from an I/O device,
OUT - to send data to an I/O device

